潘星星 

[Online Version: https://paxinla.github.io/my-online-resume/en/]

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Position：Data Engineer | Salary：Negotiable | City：Shenzhen or Remote

I have many years of professional experience in data development, familiar with the entire data processing process, including data cleaning, data storage, and solving various data problems (data backlog, data delay, data transmission efficiency, etc.). My main work areas include the design and implementation of enterprise data warehouses, ETL pipelines of massive data, design and construction of data processing platforms, design, implementation and maintenance of OLTP database architectures, SQL performance tuning, etc. I efficiently collaborate and coordinate in projects. I have designed and implemented complete data warehouse projects for a china large commercial bank, as well as implemented Big Data solutions for Internet companies.

The industries I have worked in include traditional IT, Internet, and AI companies. I prefer working with teams that are rigorous and practical. I hope to join a company that has long-term business development plans and is committed to creating continuous value for society.

TECHNOLOGY STACK

Practiced

- Python/Scala : ★★★★★
- OLAP / ETL : ★★★★★
- OLTP : ★★★★☆

Competent

- Prefer Programming Language: SQL, Bash, Clojure
- Familiar: PostgreSQL, Spark/Hadoop, RabbitMQ, MongoDB, Kettle

CERTIFICATE

- PMP | Project Management Professional
- OCP 11g | Oracle Certified Professional 11g
- Software Design Engineer | Qualification Certificate of Computer and Software Technology Proficiency

EDUCATION

- 2008~2012 Chongqing University of Posts and Telecommunications Computer Science and Technology - Bachelor
- 2008~2012 Chongqing University of Posts and Telecommunications Communication Engineering - Bachelor minor

WORK EXPERIENCE

1. Starboard Networks Pte. Ltd. -- Data Engineer , 3 year(s), Since 2021
2. Badou Intelligent Technology (Guangzhou) Co., Ltd. -- Data Engineer , 1 year(s), Since 2019
3. Shenzhen Aipin Information Technology Co., Ltd. -- Data Engineer , 3 year(s), Since 2015
4. Shenzhen Forms Syntron Information Co., Ltd. -- Dev DBA , 3 year(s), Since 2012

PROJECT

【Oct 2021 - Jan 2025】 A Cloud-Native Data Lakehouse Project

- Data Lakehouse | Spark/Databricks | Great Expectations

`This project involved the establishment and continuous enhancement of a large-scale lakehouse system, providing robust data support for the company's research teams and various Web3 products.`

My responsibilities in this project included:

1. Designing, planning, and implementing modern lakehouse and service facilities tailored to the company's business needs.
2. Analyzing and constructing efficient data models and ETL processes, continuously building and improving high-performance data services.
3. Participating in data research and analysis to meet product data requirements and support business decision-making.
4. Summarizing and maintaining technical documentation to promote the development of high-quality products.

The enterprise-grade lakehouse system under my purview is built on a cloud-native solution powered by Databricks on AWS. It leverages Databricks to handle the entire data workflow—from exploratory research all the way to production-grade workloads—while employing Great Expectations for rigorous data quality validation and monitoring. This lakehouse completely breaks free from the legacy Hadoop ecosystem, delivering vastly superior performance in managing diverse data formats and processing data at scale. Crucially, the architecture significantly reduces the company's physical infrastructure and operational overhead without increasing the risk of vendor lock-in. Using this system, I integrate and process raw data from the blockchain and smart contracts—amounting to roughly one million rows per day—and efficiently provide high-quality, reliable data to various Web3 data product teams and data scientists across the organization.

【Aug 2016 - Jan 2019】 Manage Production OLTP databases

- PostgreSQL | Alibaba Cloud

`This project serves as the online database for the company's core product, "X志愿" and is deployed on Alibaba Cloud. It reliably supports the data management needs of the product.`

My responsibilities in this project included:

1. Responsible for the design, implementation, daily operation, configuration, monitoring, and troubleshooting of the database system architecture across all environments, including development, testing, and production. In charge of monitoring backup status, developing recovery strategy plans, and creating and executing data backup and recovery plans.

2. In charge of database performance monitoring and tuning, promptly identifying issues and optimizing the way the backend accesses the database, as well as addressing poorly performing SQL queries.
3. Responsible for the logical structure design of databases in projects. This involves analyzing business requirements, designing, reviewing, and creating database objects, as well as maintaining them.
4. Generate and update core datasets required by products in bulk based on business needs, ensuring their quality is maintained.
5. Quickly locate and resolve database-related faults and performance issues.

"X志愿" is one of the company's core data products. It relies on historical enrollment data from colleges and universities nationwide, combined with AI algorithms, to predict the current year's admission outcomes for target school majors based on the personal circumstances of college entrance exam candidates. It provides intelligent and personalized advice for filling out volunteer applications. The company is one of only three volunteer application service providers in the Baidu App nationwide.

The company uses Alibaba Cloud as its cloud service provider, and all production environments are deployed on Alibaba Cloud. I am fully responsible for the design and implementation of the database architecture for this product.

After evaluating the performance and stability of Alibaba Cloud RDS, the company considered various factors such as price, data access characteristics of the "X志愿" product, and the overall product's requirements for database performance and stability. As a result, the decision was made to build a production database cluster based on PostgreSQL on ECS. I designed and implemented the database cluster architecture solution, which has been operating stably without any failures throughout the year. Compared to the previous generation of MongoDB-based database solutions, the new cluster offers significant improvements in read/write performance, development flexibility, feature richness, efficiency of batch data updates, and service stability. I have also automated the backup strategy implementation for this cluster, ensuring the security of product data.

The access to the database by the business system and the growth of basic business data experience a significant surge during the period around the college entrance exams every year, while the workload remains relatively stable during other periods. Investing resources to handle such peak access at all times would undoubtedly be a huge waste. Therefore, I have designed a solution that allows the cluster to smoothly scale up and down before and after business peak periods. During peak hours, the cluster stably supports approximately 8k TPS, while outside of peak hours, it efficiently supports the business in a cost-effective manner.

【Nov 2015 - Nov 2018】 A data warehouse in the field of job seeking and employment

- Data Warehouse | Hadoop | PostgreSQL

`This project serves as one of the company's data infrastructure components, providing stable and high-quality data support as well as technical support for massive data processing to data scientists and data mining engineers in the AI team.`

My responsibilities in this project included:

1. Responsible for the construction of the core data warehouse and data marts in the Big Data platform, including development standards, dimensional modeling, and quality metric specifications.
2. Lead data development team members in processing data, including data cleaning, quantification, integration, storage, etc.
3. Plan and implement technical solutions for the Big Data platform.
4. Provide technical support to the AI algorithm team for processing massive amounts of data.

This project is a crucial component of the company's basic data infrastructure, integrating years of data in the field of job seeking and employment acquired by the data collection team. Its primary purpose is to provide datasets for data scientists and data mining engineers in the company's AI algorithm team, and to apply their generated models to the existing data (approximately 30TB) to obtain results for use in the company's data products. Additionally, it periodically generates employment statistics reports for colleges and universities in batches, which are utilized by the company's knockout product, "XX志愿".

In this project, I led the data development team to design and implement a data warehouse solution based on CDH and PostgreSQL, aligned with the company's data strategy objectives and the unique characteristics of employment-related data. We utilized various components such as Pentaho Data Integration, DataX, Hive, MapReduce, and Spark to build ETL processes and data pipelines. Additionally, we constructed a job scheduling system based on Apache Airflow. This centralized, automated scheduling and monitoring system replaced the previously scattered and reactive data transformation jobs that the company had been using. As a result, a full-scale computation job that originally took approximately one week to complete was reduced to just 1.5 days on the new platform.

【Jun 2013 - Aug 2015】 A Business Data Loading Platform for a commercial bank

- Orchestration | Massive Data ETL

`This project is a platform for implementing ETL processes and scheduling for various characteristic application systems of the Shenzhen Branch of a large commercial bank. It customizes, processes and delivers data from heterogeneous data sources to the online library of the business system according to the needs of various downstream business systems.`

My responsibilities in this project included:

1. Responsible for maintaining and developing the orchestrating and core job procedures of the platform.
2. Design and maintain the ETL processes for data files transferred from the head office and data in the specialized system database on the platform. Handle any platform failures promptly.
3. Customize ETL processes according to new requirements raised by users of different specialized application systems, monitor and ensure the successful completion of daily batch processing.
4. Assist in retrieving historical data for specialized application system users as required.

The project involves multiple heterogeneous data sources belonging to other organizations' systems, and the destination is the application-level data marts and OLTP databases of various departmental specialized business systems. There are certain requirements for the execution efficiency of batch jobs. After me taking over the project, I completed the transformation of most of the existing historical job programs, improving their fault tolerance and the platform's ability to automatically handle errors.

For example, the platform has a source table with approximately 40 million account records, which serves as the data source for many downstream target systems. On quarterly interest settlement days, the significant increase in data volume often led to excessively long runtimes and frequent failures of the loading job (in merge mode), requiring manual on-site monitoring and intervention. After me taking over the project, I optimized the ETL process for this table by automatically splitting and slicing the source data files, analyzing incremental changes, and merging them into the existing snapshot table. This improvement enhanced the overall data loading efficiency and automation level, ensuring that the job runs stably and normally both on interest settlement days and regular days.

【Jul 2013 - Feb 2014】 A financial regulatory reporting system for an overseas branch of a commercial bank

- Data Warehouse

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This project is the construction of a regulatory reporting data warehouse for an overseas branch of a large China commercial bank located in Europe. It is capable of generating regulatory reporting data files in accordance with the data and format requirements of the local regulatory authorities. It also produces various types of user reports in a fixed format for the branch's customers and provides internal analysis reports for the branch's business departments.
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I participated in the complete process of this project, from requirement analysis to deployment and maintenance. The tasks that I was responsible for include:

1. During the requirements analysis phase, I assisted the project manager in communicating with the business department, clarifying requirements, planning metadata, and participating in the compilation of all project documents.
2. During the development and testing phase, I designed table structures and data processing programs, and completed the development and testing work.
3. During the deployment and implementation phase, I independently supported the deployment of project versions at the data center site, and developed and implemented data backup strategies for operations engineers.
4. During the maintenance phase, I provided remote technical support services.

In the project, our data team discovered that the structure of the backend data of the Murex system used by the overseas branch is significantly different from the structure of other systems in the bank, requiring a lot of integration work. In addition, the overseas branch has many business types that are not found domestically, making the logic of handling bank transactions more complex than before. Through the collective efforts of me and my team colleagues, we clarified the conversion processing rules for the business areas of concern to the overseas branch before the first phase of SIT testing in the project, and clearly defined the scope of requirements. We performed multiple processes on the backend data of their purchased funds processing system, ultimately enabling the smooth integration of regulatory reporting data for this business area and meeting the calculation requirements of all related reports. Our technical capabilities and data quality have been recognized by the overseas branch.

During the project deployment phase, I, as a representative of the data development team, participated in the deployment process at the headquarters' Beijing data center along with the entire development team from Shenzhen. During the system rollout by the headquarters' operations team, there was an issue with the production environment that caused the related batch scheduling system to fail to start smoothly. As the only member of the data development team present at the scene, under the pressure of approaching deployment time and the presence of the headquarters' team leaders, I successfully analyzed and identified the cause of the problem and promptly resolved it, ensuring the timely and normal operation of the system.

【2013】 A Private Customer Historical Transaction Query System of a commercial bank

- ETL | Oracle

`This project is a historical transaction query system for private customers of a Shenzhen branch of a large state-owned commercial bank. It integrates historical transaction data from heterogeneous data sources to meet the transaction history query needs of the bank's counter and retail departments for private customers in all branches.`

My responsibilities in this project included:

1. Analyze business requirements and perform modeling and design in Oracle databases.
2. Develop ETL programs to clean, transform, migrate, and load historical transaction data from heterogeneous data sources into the database of this system.
3. Develop ETL programs to clean and load daily incremental transaction data into the database of this system.
4. Develop and optimize query interface services for the system's backend database querying.

This project is a query application for transaction data for all individual customers of the branch. The transaction data has gone through three generations of core systems, so the structure of the transaction data differs significantly between different generations. Historical data is exported from backup files in the tape library, while newly generated data is in the form of customized structured files periodically issued by the head office. At the time of my involvement in this project, there were approximately 1.5 billion records of existing data, growing at a rate of 1 million per day.

My main roles in the project include conducting an initial analysis of the raw data structure for the 3rd generation system, standardizing the new structure, and modeling the data tables. When dealing with historical tape transaction data files (approximately 20 years), I developed programs to parse and transform the data, and loaded it into the database, completing the task in about two days. In the later stages of the project, I optimized the query interface services by integrating them into Oracle packages for use by the backend system. Before the optimization, the query time sometimes approached seconds, but after the optimization, the average query time stabilized at the microsecond level.

Acknowledgement

Thank you very much for taking the time to read my resume. I look forward to the opportunity to work with you.